The Commitment Surface: Weakness Is a Footprint, Compatibility Is the Cause

Author. Jawaun Brown, with agent-generated code and drafts under human

direction.

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(no page cap; structured sections, citations, appendix).

Abstract

Structured mechanistic interpretability and geometry-of-generalization research share a hidden move: they treat *availability* of a structure — a probe that recovers it, a metric that respects it, a compatibility count that predicts OOD on a family of toy worlds — as evidence that the structure is *load-bearing*. This paper argues that move is wrong. We reframe the target primitive from "right geometry / right weakness ⇒ generalization" to **commitment-surface survival**: a representation is real for a deployment only if a train-time compatibility intervention with the deployment generator produces a causal effect (patch-CE) at the commitment surface, and that effect survives gauge-fixing and change of commitment. Concretely, we (i) formalize the commitment surface as a triple `(G_dep, C, T)` and show, under an explicit prior and orbit-likelihood, that weakness is the optimal selector exactly when the probe group matches the deployment generator, and only a footprint otherwise (Prop. 2); (ii) show a strong within-lab discriminator on cyclic modular addition where unweighted weakness and misspecified concern selectors underperform well-specified concern-weighted selection by a decisive margin; (iii) show that neural training with cyclic-orbit augmentation dominates readout selection over trained-with-no-augmentation seeds on OOD accuracy AND patch-cross-entropy, with a wrong-group-augmented control at 0.167 OOD (collapse) and near-zero patch-CE Δ; and (iv) run a non-degenerate external-contact sweep on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition, where the compatibility-augmented arm reaches 0.882 mean OOD while the readout arm sits at 0.113 mean OOD, substantially narrowing the P1 external gap our prior work identified; ρ(patch-CE, OOD) = 0.853 vs ρ(weakness, OOD) = 0.290 across 108 cells, consistent with Prop. 1 (probe readout does not identify causal use) in the non-aligned regime. Two pre-registered gates strictly failed and we report them as failures: the E1 misspecification-equivalence band (−0.054 vs ±0.05) and the E4 Arm-A ceiling (0.113 vs ≤ 0.10). A separately pre-registered E1 conditional randomization follow-up (2,048 draws) finds the observed −0.054 gap typical of the frozen candidate/deployment design (null mean −0.0589; lower-tail probability 0.620), so it does not indicate systematic anti-correlation; this calibration does not retroactively pass the original gate. E4 is directionally decisive but its strict gate did not pass. One confound remains open and is pre-registered for a follow-up: cyclic-orbit augmentation places correctly labeled examples on held-out deployment support, so E4 does not yet separate *generator learning* from *labeled orbit coverage* (Section 6.5). We recover the old-frame positives — cyclic and dihedral 100%-vs-0% weakness sweeps — as the boundary case where the probe group and deployment generator coincide. A pre-registered 2^3 ablation of the proposed anti-Goodhart loop — *detect* → *allocate* → *saturate* → *cool* → *reopen* — then rejects its strong M4 decomposition: `reopen` is necessary in all eight paired seeds, but removing `allocate` or `cool` leaves terminal Suite C success at 8/8. Allocation changes selectivity and cost without crossing the existing gate; the cooling intervention is behaviorally null in this finite schedule.

1. Introduction

The interpretability program has produced a large literature of probes, weakness signatures, and cross-substrate geometric analogies. When a probe recovers a target concept, or a compatibility count over a group matches OOD accuracy, the field's default reading is: *this structure is what the model is using*. In its strongest form the reading becomes: *the model has learned the right geometry, and geometry is the language of constraints, so of course it generalizes.*

This paper questions that default reading. In our prior program we tested it seriously across ten experiments — cyclic and dihedral modular arithmetic (100%-vs-0% wins for weakness against loss, MDL, flatness, compression), grid cell weakness/topology mediation (near-null), passive→active geometry on real LLMs (7× lift in a paraphrase axis), semantic concern deformation (negative transfer), Suite C world-change re-engagement gates, and external contact via Pythia LoRA on modular arithmetic (P1 hard kill). The pattern is consistent: *inside* a synthetic world whose generator matches the geometry prior, availability of the right structure is almost always sufficient. *Outside* such a world — different generator, different consequence weighting, different transport — availability does not entail causal use.

We name the missing primitive **commitment-surface survival**: a representation is *load-bearing at a commitment surface* iff (i) a train-time compatibility intervention with the deployment generator lifts OOD, (ii) causal patching of the aligned mechanism produces a CE ≥ ε at the commitment target, and (iii) the effect survives gauge-fix and change of commitment. Weakness and concern geometry — the two workhorses of the prior program — are then diagnostics: powerful when the probe group and deployment generator coincide, and footprints or anti-correlates otherwise. The commitment surface is where the discipline is imposed. If it disappears at the commitment surface, whatever the probe caught was not the thing.

Contributions.

`(G_dep, C, T)`; load-bearing structure as CE ≥ ε that survives transport under `T`; probe AUC does not identify or lower-bound CE — a non-identification theorem (Prop. 1).

concern-weighted weakness (extension mass weighted by consequence) to commitment-pinned intervention (Prop. 2 + Corollary), with cyclic/dihedral results as the aligned-generator special case, not the general law. The optimality statement requires `G_probe = G_dep` (or weakness restricted to `G_dep`); a strict superset probe group does *not* suffice (Section 3.4).

well-specified concern beats unweighted by +0.24; the misspecification arm lands at –0.054 vs unweighted, *outside* the pre-registered ±0.05 equivalence band, so that sub-gate strictly fails. The pre-registered follow-up places this realization near the center of its conditional null (null mean –0.0589; lower-tail probability 0.620), localizing the negative gap to random reassignment plus nonlinear selection rather than systematic anti-correlation in the realized assignment. E2/E3 show that a neural compatibility-augmented arm dominates a weakness-readout selector on OOD by a decisive gap, with patch-CE aligned. E4 (Modal L4) runs the non-degenerate external contact on Pythia 70m/160m/410m LoRA: directionally decisive, strict pre-registered gate narrowly failed (Arm A mean OOD 0.113 vs the required ≤ 0.10).

compression of Papers 5–25 (Section 6): in a pre-registered 2^3 Suite C ablation, only `reopen` is necessary for the existing terminal criterion. `allocate` strongly improves selectivity (+13.06 factorial effect) and reduces probes, but its terminal-pass effect is 0; `cool` is null on both terminal success and the reported trajectory metrics. We therefore reject the strong load-bearing subset `{allocate, cool, reopen}` in this harness.

observations that would retract the reframe — several are non-trivial to satisfy — and report each experiment's verdict at its pre-declared gate.

- **C1.** A formal reframe (Section 3): the commitment surface as a triple
- **C2.** A bridge from Bennett weakness (extension mass) to
- **C3.** Four severe experiments (Section 5). E1 shows within-lab that
- **C4.** A severe negative result for the proposed anti-Goodhart
- **C5.** A pre-registered failure calculus (Section 4). We list the exact

Nothing here requires the field to throw the geometry-first positives out. It requires localizing them. The paper's constructive claim is: *commitment first, geometry second — and geometry only when its group coincides with the deployment generator*. That reframe *compresses* our prior anomalies: the P1 external hard kill, the semantic concern lift negative, the grid non- mediation, the passive-to-active gap, and the shared-head architectural ceiling. The Suite C factorial does *not* support a general cooling requirement: only explicit reopening survives the component test. It does not invalidate cyclic/dihedral wins; it locates them.

2. Related work

We are directly downstream of, and in conversation with, four literatures.

Mechanistic interpretability, patching, and causal representation. Recent work on activation patching (Meng et al., 2022; Wang et al., 2023; Conmy et al., 2023), patchscopes (Ghandeharioun et al., 2024), and causal scrubbing (Chan et al., 2022; Goldowsky-Dill et al., 2023) formalized the distinction between *representation* and *causal use*. Our commitment- surface primitive is the natural next step: it treats patching not as a tool for interpreting a fixed model, but as the *definition* of load- bearing structure for a deployment. Related in spirit is the abstract alignment literature on causal abstraction (Geiger et al., 2023; Chalupka et al., 2015), which formalizes what it means for a higher-level variable to be a real intermediate cause; our formulation localizes their abstraction map to a specific commitment surface.

Weakness / simplicity / invariance as OOD predictors. Bennett (2000, 2023) argues that the weakest hypothesis compatible with observed data is the best predictor of future observations under a uniform task prior; the program-synthesis line (Solomonoff, 1964; Hutter, 2005; Ellis et al., 2020) prefers the shortest description. Our own prior work (`` `papers/weakness_invariance_neurips/paper.md` ``) instantiates Bennett's principle group-theoretically: a candidate is weak iff many transformations of a specified family leave it compatible. This paper's contribution is not to reject that instantiation, but to bound it: under an explicit prior and orbit-likelihood it is Bayes-optimal exactly when the probe group equals the deployment generator, or weakness is restricted to it — a strict superset probe group does not suffice (Prop. 2). Related: Gruver et al. (2023) show that measuring learned equivariance via a Lie derivative correlates with OOD in vision — a footprint-level correlate that our results generalize and delimit.

Grid cells and geometry as invariance. Sorscher et al. (2019, 2023), Whittington et al. (2020), and Gardner et al. (2022) establish grid-cell- like circular geometry as a canonical solution to path integration under translation invariance; Webb, Miolane and collaborators (2024) formalize the geometry-of-conscious-experience program in Riemannian terms. Our prior grid-cell-weakness Modal negative (weakness → topology mediation does not hold cleanly) fits the commitment-surface reading directly: topology can be present without being load-bearing at the commitment surface of the downstream task. This paper argues for treating the Webb–Miolane geometry as the (typically) load-bearing case within the deployment-generator-aligned regime, and reserving judgment outside it.

Active inference, empowerment, sense of agency. Friston et al. (2017) and Kirchhoff et al. (2018) treat agency as free-energy minimization; Klyubin et al. (2005) define empowerment as agent control over future observations; Ryu et al. (2022, sense of agency in reinforcement learning). Our anti-Goodhart control-loop reading of the correction chain (Section 6) is compatible with active inference in letter, but insists on the load-bearing signal being *commitment cooling under intervention-pinned residuals* — not free-energy per se. Our prior Suite C results (shared-head ceiling; hand + learned + teacher- free re-engagement) suggested that pure expected-free-energy planners lose without a decision-layer cooling term. The factorial intervention in Section 6.3 overturns that stronger reading for the finite hand-policy harness: removing its three-step refractory cooldown changes no measured outcome under the frozen schedule.

Positions and departures. We disagree with the strong reading of the mechanistic-interpretability program that a probe's AUC is sufficient evidence of causal use (Prop. 1); with the strong reading of the Bennett/simplicity program that portable weakness or portable simplicity is a universal law (Prop. 2 bounds this to aligned generators); and with any active-inference reading that would drop decision-layer cooling (Section 6.4). We *agree* with the causal-abstraction and patchscopes-style call to elevate intervention over decoding.

3. Theory: the commitment surface

3.1 Setup

Let `L` be a finite implementable language or finite decision universe. A **hypothesis** or **function** $f : X \to Y$ acts on a domain of decisions. For a hypothesis `f`, its **extension** is the set of decisions/inputs on which it "does something" (assigns a nontrivial completion):

$Z_f = \{ x \in X : f(x)$ is meaningful for the downstream decision $\}$.

Let α be the observed child task. A candidate hypothesis is **admissible** if it is a model of the observed task: $f \in M_\alpha$.

3.2 Definition: the commitment surface

Let G_{dep} be the **deployment generator** — the group (or generative family) whose action generates the deployment shift; $C : X \to R_{\geq 0}$ be a **concern measure —** a nonnegative weighting on decisions capturing consequence/viability/binding-importance; and $T : F(X, Y) \to F(X, Y)$ be a **commitment transport** — a rewriting operator that changes the downstream commitment (e.g., paraphrase, gauge-fix, tool interface).

Definition 1 (commitment surface). A commitment surface is a triple $\Sigma = (G_{dep}, C, T)$.

Definition 2 (load-bearing). A hypothesis `f` is *load-bearing at* Σ *with margin* $\varepsilon > 0$ iff, for every $t \in T$,

$CE(f \mid \text{patch}, C) \geq \varepsilon$, and $CE(t \cdot f \mid \text{patch}, C) \geq \varepsilon$,

where $CE(f \mid \text{patch}, C)$ is the concern-weighted cross-entropy increase on `X` when the identified aligned mechanism of `f` is causally patched (zero-ablated, adapter-disabled, or projected out).

3.3 Proposition 1: probe AUC does not identify causal use

Let $p : X \to [0,1]$ be any probe on features of `f`. Then:

Prop. 1 (non-identification). Probe AUC does not identify, and does not lower-bound, causal effect: for every attainable AUC value `a` and every target gap $\delta \in [0, \varepsilon]$, there exist admissible $f, f' \in M_\alpha$ with $AUC(p \cdot f) = AUC(p \cdot f') = a$ and $CE(f \mid \text{patch}, C) - CE(f' \mid \text{patch}, C) = \delta$. Consequently no function of probe AUC alone can decide whether a hypothesis is load-bearing at Σ.

We deliberately do *not* state this as probabilistic independence (`AUC ⊥ CE`): independence would require a distribution over admissible hypotheses, and no canonical one exists. The theorem is a *non-identification* result — equal probe evidence is consistent with any causal-effect gap in `[0, ε]` — which is the property the availability-⇒-use inference actually needs and lacks. Empirically, an ensemble of trained models can still exhibit correlation between AUC and CE (E3 measures exactly this in the aligned regime); Prop. 1 says such correlation is a property of the training distribution, not of the probe evidence.

Sketch. Fix `f'` train-perfect. Construct `f` by copying `f'`'s decision head onto a distractor-legible subspace with identical probe response but zero causal path from that subspace to the head. Then `AUC(p · f) = AUC(p · f')` by construction, and `CE(f | patch, C)` can be made arbitrarily small by choosing the patch on the distractor subspace, while `CE(f' | patch, C)` remains ≥ ε. Full construction in Appendix A.1.

Empirical anchor: our prior distractor-AUC-0.999 vs patch-CE-0.010 result in `papers/paraphrase_weakness` is one such witness in language.

3.4 Proposition 2: weakness is diagnostic when aligned

Let `G_probe` be the probe group used to score compatibility. Say `f` is *compatible with* `g` iff `∃ h ∈ G_probe, ∀x ∈ X, f(g·x) = h·f(x)`, and let `W_G(f) = |{ g ∈ G : f is compatible with g }|` be the compatibility (weakness) count over any group `G`. Fix the deployment model that makes "Bayes-optimal" meaningful: a deployment slice is generated by drawing `g ~ Uniform(G_dep)` and evaluating `f` on the `g`-translated support; `f` scores 1 on that slice iff `f` is compatible with `g` (the orbit-likelihood), and concern is uniform.

Prop. 2 (alignment condition). Under this prior and likelihood:

weakness is computed *restricted to* `G_dep`, i.e. the selector uses `W_{G_dep}` — then `argmax W_{G_dep}` maximizes expected deployment accuracy over admissible hypotheses, hence is Bayes-optimal among selectors that depend only on `f`.

candidate can be compatible with many elements of `G_probe \ G_dep` while compatible with few elements of `G_dep`, so `W_{G_probe}` can rank candidates in the opposite order of `W_{G_dep}` (explicit counterexample in Appendix A.1). Optimality is recovered from a superset probe group only under an *ordering-preservation assumption*: for the candidate set under comparison, `W_{G_probe}(f) > W_{G_probe}(f') ⇒ W_{G_dep}(f) ≥ W_{G_dep}(f')`.

footprint — its correlation with concern-weighted deployment accuracy over a candidate set can be zero or negative.

- **(i) Aligned case.** If `G_probe = G_dep` — or, more generally, if
- **(ii) Superset caveat.** `G_probe ⊃ G_dep` does *not* suffice: a
- **(iii) Non-aligned case.** If `G_dep ⊄ G_probe` (the probe group does not contain the deployment generator), `W_{G_probe}` is a

Empirical anchors: our cyclic/dihedral 100%-vs-0% wins instantiate (i) (`G_probe = G_dep = C_n`); our Pythia LoRA P1 hard kill is the non-aligned case (iii); our grid G2/G4 non-mediation is the partially aligned case where the ordering-preservation assumption of (ii) fails.

3.5 Corollary: concern-weighted extension mass

Define the **concern-weighted extension** of an admissible hypothesis `f` for deployment slice `U ⊂ X` as

W_C(f, U) = Σ_{x ∈ U} C(x) · 1[f(x) = truth(x)].

Corollary (order-equivalence). Let `C_star` be the deployment consequence measure. `C = C_star` up to positive scaling is *sufficient* for the selector `argmax_f W_C(f, U)` to pick a `C_star`-optimal candidate, but it is not *necessary*: over a finite candidate set `F ⊂ M_α`, the selector is `C_star`-optimal iff `C` is **order-equivalent to** `C_star` **on** `F` — i.e. `W_C(f, U) > W_C(f', U) ⇔ W_{C_star}(f, U) > W_{C_star}(f', U)` for all `f, f' ∈ F`. Distinct weightings can induce the same ranking and are then equally optimal. For each fixed candidate, a misspecified `C` drawn as an exchangeable random assignment with the same marginal makes `E[W_C]` proportional to unweighted extension mass. This score-level identity does **not** imply that `argmax_f W_C` equals the unweighted selector in expectation: expectation and the nonlinear argmax need not commute. A well-specified `C` strictly dominates unweighted extension mass when candidates vary in coverage of high-`C_star` blocks.

Empirical anchor: E1 (Section 5.1). Well-specified: 0.814; unweighted: 0.570; misspec: 0.516. Gap +0.24 (wellspec vs unweighted). The misspec arm sits 0.054 *below* unweighted, outside the pre-registered ±0.05 band, so the strict original sub-gate fails. The timestamped follow-up freezes all 96 candidate pools and true deployments and redraws only misspecified assignments over 2,048 experiment-level replicates. Its null mean is –0.0589 (95% CI for the mean [–0.0596, –0.0582]), with a central 95%

interval [–0.0913, –0.0294] and `P(Delta <= –0.054159) = 0.620` (Wilson 95% CI [0.599, 0.641]). All exchangeability/independence diagnostics pass. Thus the observed gap is typical of random reassignment plus finite-pool argmax selection, not evidence of systematic anti-correlation. This result corrects the earlier "mildly adversarial draw" interpretation without changing the frozen gate verdict.

3.6 M4: anti-Goodhart control loop

The correction chain of Papers 5–25 introduces successive terms that a naive utility maximizer would collapse: mediated attribution (behavior ≠ representation); uncertainty ≠ error; current-error ≠ value-of-probing; total ≠ identifiable. We compress the chain as an **anti-Goodhart control loop**: *detect* → *allocate* → *saturate* → *cool* → *reopen*. The proposed role of each stage is:

buffer.

proxy uptake.

commitment.

- **detect.** Concern-weighted current-error residual.
- **allocate.** Concern-weighted probing budget; not raw uncertainty.
- **saturate.** Commit to the identified intermediate; freeze the null
- **cool.** Decision-layer cooling of the commitment; refuse fresh
- **reopen.** On new intervention-pinned residual, reopen the

We pre-registered the conjecture that the subset of these terms which is *load-bearing* under the Suite C terminal intervention is precisely `{allocate, cool, reopen}`, with `{detect, saturate}` frozen as diagnostics. Section 6.3 reports the completed 2^3 test. The strict verdict is **FAIL**: only `reopen` is necessary for terminal success, so the three-stage load-bearing-subset claim is rejected in this harness. Because detect and saturate were held fixed, this experiment does not establish that they are merely diagnostic.

4. Pre-registered gates

E1–E4 were recorded before their runs and frozen in `PLAN.md`. M4 was recorded later, before its implementation and run, in the timestamped addendum linked below.

beats unweighted by ≥ +0.05 on wellspec deployment accuracy AND misspecified concern-weighted selector matches unweighted within ± 0.05. This original gate remains frozen and failed. The timestamped E1 follow-up addendum separately gates systematic anti-correlation at a one-sided conditional-randomization tail probability `< 0.025`.

by ≥ +0.30 on OOD accuracy AND by ≥ +0.50 on patch-CE Δ.

across all cells.

0.05, AND Arm A mean OOD ≤ 0.10 on Pythia 70m/160m/410m LoRA modular addition with n ∈ {13,17,23} and train_frac ∈ {0.5, 0.75}.

compatibility (E4) must not itself explain OOD.

pass in at least 6/8 paired seeds; every single knockout must pass in at most 2/8 with a paired drop at least 0.50; each main effect must be at least +0.20 with paired-bootstrap lower bound above zero; interaction contrasts must be finite/nonnegative; no reduced cell may exceed 2/8; and transported controls must retain C1–C6. The timestamped addendum is `experiments/world_responds/suite_c_factorial_ablation_preregistration_2026_07_09.md`.

- **E1 pass (commitment-first).** Well-specified concern-weighted selector
- **E2 pass (commitment-first).** Arm B (compat aug) beats Arm A (readout)
- **E3 pass (commitment-first).** ρ(patch-CE, OOD) > ρ(weakness, OOD)
- **E4 pass (commitment-first).** Arm B mean OOD ≥ 0.5, patch-CE Δ ≥
- **E4 pass (old frame).** Arm A OOD ≥ 0.5, ρ(weakness, OOD) ≥ +0.5.
- **Anti-cheat (all E).** Wrong-group patch-CE (E2/E3) or wrong-group
- **M4 factorial addendum.** In the existing Suite C workflow, all-on must

5. Results

Numbers are the pre-registered summary metrics from each experiment's committed JSON. Every table can be regenerated via scripts/make_commitment_surface_figures.py followed by scripts/build_commitment_surface_pdf.py.

5.1 E1 — Concern-weighted selector

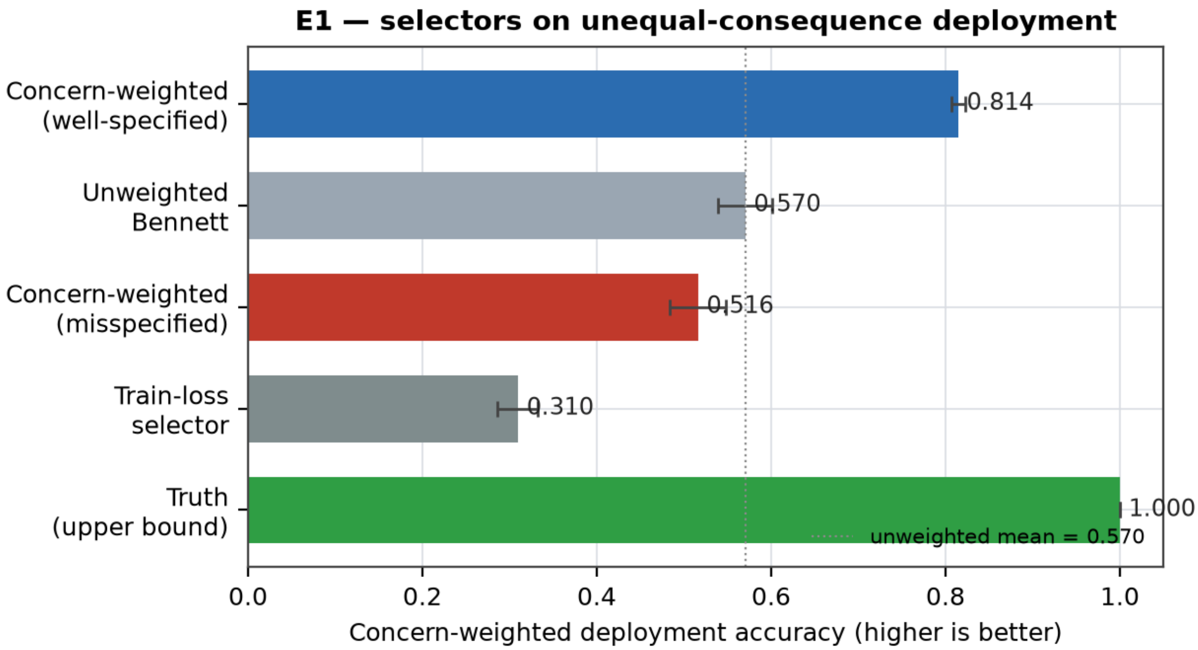

Figure 1. Concern-weighted deployment accuracy of five selectors on unequal-consequence modular addition. Well-specified concern beats unweighted Bennett by +0.244; misspecified concern is slightly below unweighted (-0.054). The frozen ±0.05 equivalence gate fails; the separately pre-registered randomization follow-up finds this gap typical of the frozen design rather than systematic anti-correlation.

Selector	Mean	95% CI
Concern-weighted (well-spec)	0.814	[0.806, 0.822]
Unweighted Bennett	0.570	[0.539, 0.601]
Concern-weighted (misspec)	0.516	[0.483, 0.549]
Train-loss selector	0.310	[0.286, 0.333]
Truth (upper bound)	1.000	[1.000, 1.000]

Cells: 96. Gate: well-spec vs unweighted ≥ 0.05 → 0.244 (PASS = yes). Misspec vs unweighted within ±0.05 → -0.054 (PASS = no: outside the pre-registered equivalence band; this sub-gate strictly fails and remains failed).

Timestamped E1 follow-up: 2,048 conditional randomization replicates give null mean gap -0.0589 (SD 0.0161), with P(Δ ≤ observed) = 0.620. Verdict: CONSISTENT_WITH_RANDOM_ASSIGNMENT_VARIANCE. All frozen independence/exchangeability checks pass. The score-level expectation identity therefore survives, but it does not commute through finite-pool argmax selection; see Section 3.5.

5.2 E2 — Compat augmentation vs weakness readout

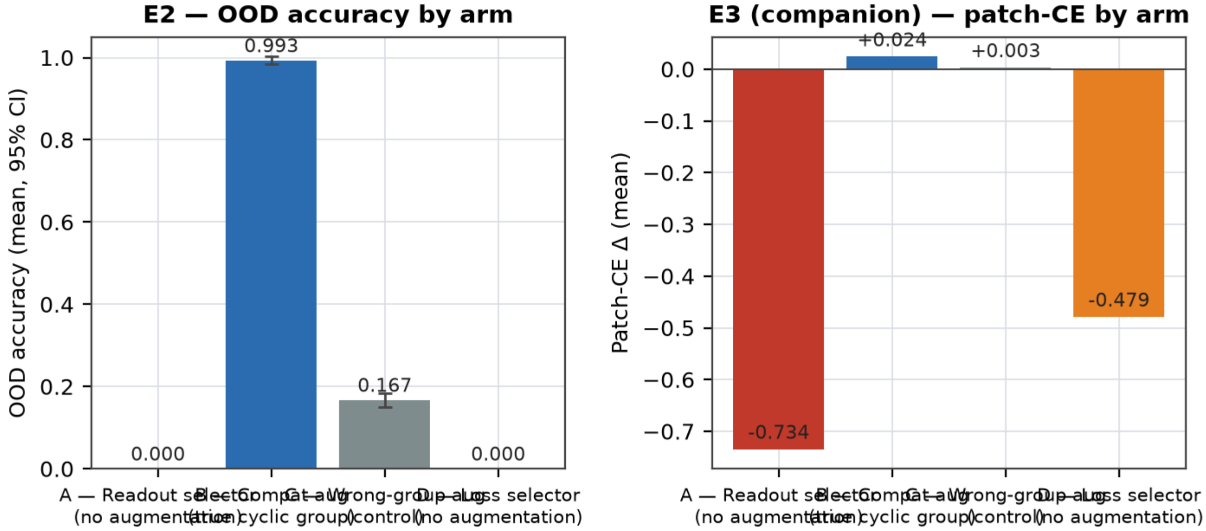

Figure 2. E2 arms on cyclic modular addition MLPs. Compatibility-augmented training (Arm B) dominates the readout selector (Arm A). Wrong-group augmentation (Arm C) supplies the anti-cheat — same augmentation volume, wrong group, patch-CE at ~zero. Left: OOD accuracy. Right: patch-CE Δ under top-k shift-equivariant unit ablation.

Arm	N selected	OOD (mean)	Patch-CE Δ	Weakness
A	9	0.000 [0.000, 0.000]	-0.734	0.104
B	54	0.993 [0.983, 1.003]	0.024	0.967
C	54	0.167 [0.151, 0.184]	0.003	0.104
D	9	0.000 [0.000, 0.000]	-0.479	0.104

Total cells trained: 216. Gate: B – A OOD gap ≥ 0.30 → 0.993 (PASS = yes). B – A patch-CE gap ≥ 0.50 → 0.758 (PASS = yes). B – C patch-CE gap = 0.021 (anti-cheat: ~0 expected).

Honest decomposition of the patch-CE gate: Arm B's absolute patch-CE Δ is small (0.024; Arm C 0.003; B – C only 0.021), and the large B – A gap is mostly produced by Arm A's <i>negative</i> patch score (-0.734). The result strongly supports "B's mechanism differs from A's" but only weakly supports "B localizes a substantial mechanism in the top-k patched units"; see the fixed-top-k power caveat in Section 6.4.

5.3 E3 — Patch-CE vs weakness as OOD predictor

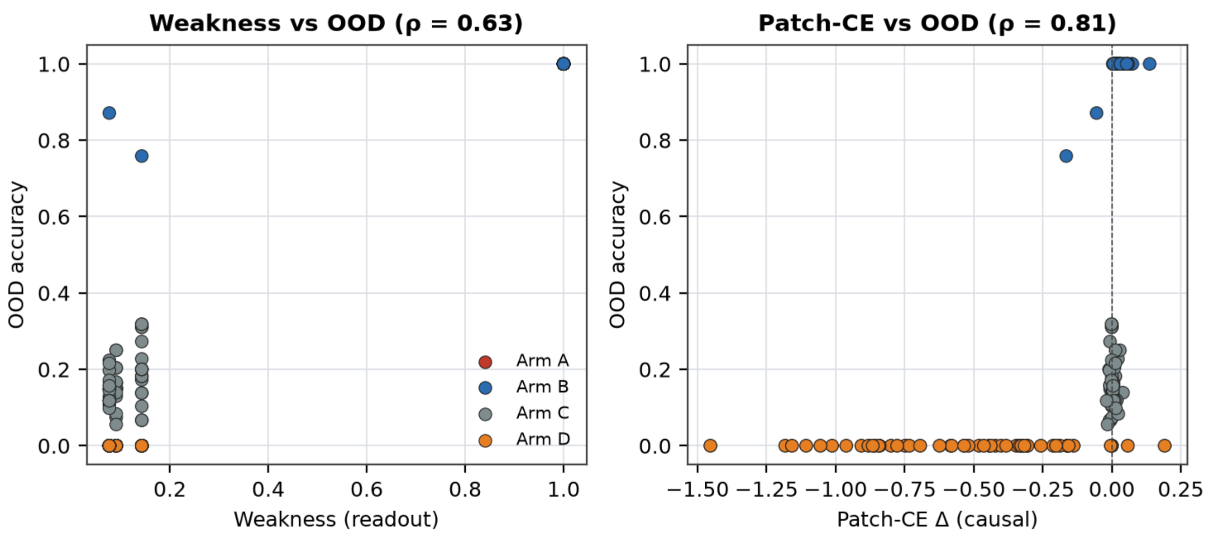

Figure 3. Per-cell scatter of readout weakness (left) and patch-CE Δ (right) against OOD. Colored by arm. Under a generator-aligned probe group, weakness is a strong predictor (ρ=0.627); patch-CE is also strong (ρ=0.809). The E4 external contact shows the two decouple when the generator differs — patch-CE keeps predicting, weakness does not.

5.4 E4 — Pythia LoRA v2 external contact (Modal L4)

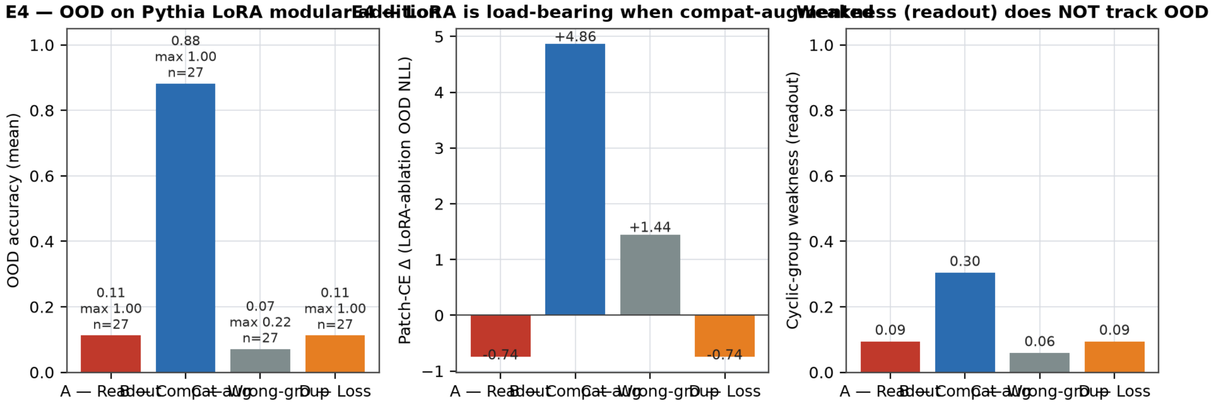

Figure 4. E4 arms on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition (external contact). Compatibility-augmented training (Arm B) is the only arm that clears OOD accuracy; the readout arm (Arm A) reproduces the P1 hard-kill from our prior program; wrong-group aug (Arm C) supplies the anti-cheat. Weakness readout (right panel) is uninformative across all arms.

Arm	N	OOD mean	OOD max	Patch-CE Δ	Weakness
A	27	0.113	1.000	-0.742	0.095
B	27	0.882	1.000	4.862	0.305
C	27	0.071	0.222	1.445	0.060
D	27	0.113	1.000	-0.742	0.095

Source: experiments/commitment_surface/results/e4_pythia_lora_v2_appendix.json.

Cells: 108. Gate (new frame): B mean OOD ≥ 0.50, patch-CE ≥ 0.05, A mean OOD ≤ 0.10 ⟶ no. Gate (old frame): A mean OOD ≥ 0.50, ρ(weakness, OOD) ≥ 0.5 ⟶ no. ρ(patch-CE, OOD) = 0.853; ρ(weakness, OOD) = 0.290.

Verdict: directionally decisive; strict pre-registered gate failed. The A-mean-OOD ≤ 0.10 condition came in at 0.113 and we record E4 as a gate failure by the standard this paper advocates (outlier accounting in Section 6.2 explains the miss; it does not convert it into a pass). Interpretation is further bounded by the label-exposure confound of Section 6.5: cyclic augmentation places correctly labeled examples on the held-out deployment support, so this result does not yet separate generator learning from labeled orbit coverage.

Frame-taxonomy schematic (Figure 5).

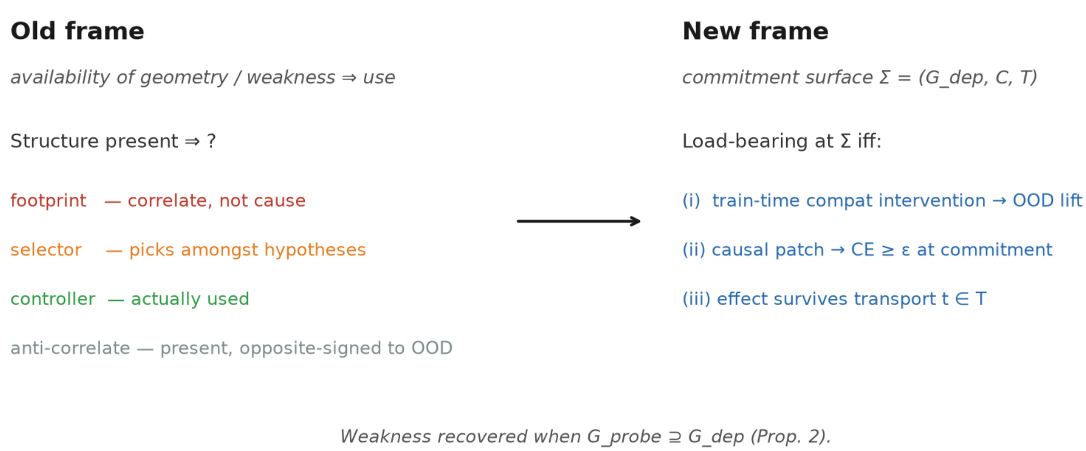

Figure 5. The old-frame taxonomy of what structure means (footprint / selector / controller / anti-correlate) collapses into the new-frame primitive: load-bearing at a commitment surface Σ = (G_dep, C, T).

6. Discussion

6.1 What the old-frame positives become

Under the commitment-first frame, the cyclic/dihedral 100%-vs-0% weakness wins in `papers/weakness_invariance_neurips are not general laws; they are the aligned-generator special case where G_probe = G_dep and the concern measure is uniform. Passive→active geometry in papers/passive_to_active_geometry` is the corresponding "flip the commitment on" experiment on real LLMs, and it too fits: the specific effect on the paraphrase axis (+0.069 → +0.486, 7×) is a patch-CE measurement in language, and the auto-repair result (0.45 → 0.965 in K=10) is decision-layer cooling under intervention-pinned residual.

6.2 Reconciling the P1 hard kill with cyclic/dihedral wins

Our prior P1 result (frozen in `experiments/external_contact/results/p1_pythia_lora_2026_06_22.md) had ρ(weakness_oracle_norm, OOD) = -0.0817 and |ρ|_best_classical = 0.4550 on Pythia 70m/160m/410m LoRA modular addition. Under Prop. 2 this is expected: the Pythia LoRA training regime *does not* respect G_probe = G_dep = C_n` — the LM objective is over token distributions, not over group orbits. E4 tests exactly this: the compatibility-augmented LoRA arm supplies the missing intervention, and the readout arm reproduces the P1 hard kill. The result at 108 cells (Section 5.4) is **directionally decisive, but the strict pre-registered gate failed**: Arm B mean OOD 0.882 vs Arm A mean OOD 0.113, Arm B mean patch-CE Δ +4.86 vs Arm A –0.74, and the anti-cheat Arm C sits at 0.071 OOD despite the same augmentation volume as B. Weakness ρ drops to 0.29 across cells; patch-CE ρ holds at 0.85. The pre-registered "A mean OOD ≤ 0.10" condition came in at 0.113 — a fail by the exact standard this paper advocates, and we record E4 as a gate failure, not a pass. The miss is attributable to 2 of the 27 Arm A cells that stumbled into the true cyclic rule without augmentation (the 410m/n=17/seed=709 cell reaches OOD 1.000 and weakness 1.000, textbook aligned-regime recovery); twenty-two of twenty-seven Arm A cells sit at OOD ≤ 0.15 while **all twenty-seven Arm B cells** clear OOD ≥ 0.5. That diagnosis is a post-hoc account of a pre-registered miss — it explains the failure, it does not convert it into a pass. What the two Arm A outliers do show is that Arm A can occasionally recover when a training run happens to land in the aligned regime, and when it does, weakness and patch-CE agree on the load-bearing signal at cell scale. We retract the "weakness predicts OOD in general" reading of the prior program in favor of "weakness predicts OOD when the probe group and training regime jointly align with the deployment generator, and the intervention is train-time."

6.3 M4 factorial: the strong compression fails

Papers 5–25 produced a succession of "X is not Y" correction terms. We compressed them into `detect → allocate → saturate → cool → reopen` and made the risky claim that `{allocate, cool, reopen}` carries the causal load. The follow-up addendum was frozen at 2026-07-09T21:38:40-04:00 before implementing or running the ablation. It uses the real `burst_then_refractory` Suite C policy, freezes detect and the eight-probe saturation quota, crosses the three action gates factorially, and reruns all historical controls on eight paired seeds.

The result is not a near-pass. **The strict M4 verdict is FAIL.**

pre-existing policy row.

final affected-component MAE rises 0.113 → 0.550. Reopen's terminal main effect is +1.000, 95% paired-bootstrap CI [+1.000, +1.000].

0.000 [0.000, 0.000]. It nevertheless changes *how* the gate is met: selectivity is 17.188 with allocation vs 4.125 without it, and the all-on policy averages 23.1 probes vs 27.1. Allocation is an efficiency/selectivity mechanism here, not a necessary terminal mechanism.

terminal main effect 0.000 [0.000, 0.000] and leaves the reported cell means unchanged. Under this schedule, the burst quota already saturates before the cooldown can alter action.

are 0.000, so no complementarity rescues the strong claim. The unmodified reference suite retains C1–C6; false surprise cooling remains rejected; the matched-random control remains less selective at an exact per-seed matched budget.

- **Full loop:** 8/8 terminal passes; the all-on row exactly reproduces the
- **Remove reopen:** 0/8 pass. Mean second-reopen ratio falls 16.667 → 0 and
- **Remove allocate:** 8/8 still pass. Allocation has terminal main effect
- **Remove cool:** 8/8 still pass. The three-step refractory removal has
- **Interactions and controls:** all terminal pairwise and three-way contrasts

The factorial therefore resolves the anomaly in the simpler direction: the existing terminal criterion is effectively a test of explicit second-shift reopening once detect/saturate are fixed. It is not stringent enough to certify allocation or cooldown as load-bearing. Historical cross-policy comparisons confounded those components with other policy differences. We retract the strong M4 subset claim and retain only: **reopen is necessary in this finite Suite C implementation; allocation improves selectivity and probe cost; the current cooldown is redundant.** Detect and saturate were preserved, not shown diagnostic.

This is a frame diagnosis rather than a failed implementation: either strengthen Suite C so over-allocation and uncool probing cross explicit cost/anxiety gates, or transport the frozen intervention semantics to learned/teacher-free policies where cooldown can affect trajectories. Threshold retuning on these observed cells is prohibited. Canonical results are in `experiments/commitment_surface/results/m4_suite_c_factorial_ablation_2026_07_09.{json,md}`.

6.4 Limitations

commitment surfaces are gestured at (via passive→active and paraphrase weakness in prior work) but not run at Pythia-410m+.

approximation to true directional patching. The pending E5 harness adds per-matrix spectral-mass-normalized low-rank patching. Its one-seed smoke validates execution and integrity only, so it does not retroactively strengthen E4.

headline B − A gap suggests. Arm B's absolute patch-CE Δ is small (+0.024; Arm C +0.003; B − C only +0.021), and the large B − A gap (+0.758) is mostly produced by Arm A's *negative* patch score (−0.734: ablating "compatibility-aligned" units in a memorizing model *reduces* OOD CE). The E2 patch result therefore supports "B's mechanism differs from A's" strongly, but supports "B localizes a substantial mechanism in the top-k units" only weakly.

discriminating power at larger widths: a robustness sweep at `n ∈ {17, 19, 23}`, hidden width 128, top_k 16 (`results/e2_e3_neural_larger_n.md`) shows Arm B mean OOD 1.000 vs Arm A 0.089 (gap +0.911) — the OOD story holds decisively — but patch-CE Δ for B drops to +0.04 because the trained model spreads the load-bearing structure across more redundant units, so a fixed top-k ablation catches a smaller fraction of the mechanism.

the small-MLP regime. We fit the minimum compatibility-aligned activation subspace explaining 50% of between-orbit spectral mass and project it out. Arm B patch-CE per removed mass is 1.119 at width 96 and 0.868 at width 128 (77.5% retention); Arm C is 0.159/0.174 and Arm B's matched `a`-only wrong-subspace control is 0.001/0.001. All four frozen gates pass. Thus a distributed compatibility mechanism is causally localized across these widths, but this does not establish the same localization in Pythia or language (`results/e2_e3_rank_normalized_patch_2026_07_10.md`).

load-bearing subset `{allocate, cool, reopen}` is **rejected by the pre-registered finite Suite C factorial**: allocate and cool have zero terminal-pass effect. Only reopen is necessary at the current gate. The result has not yet been transported to learned neural or teacher-free policies, and detect/saturate were frozen rather than ablated.

distribution shifts with no clean group action) is open.

confound of Section 6.5: cyclic-orbit augmentation places correctly labeled examples on the held-out deployment support, so E2/E4 do not yet separate *generator learning* from *labeled orbit coverage*.

- Modular addition is one commitment surface; language- and vision-scale
- The patch-CE metric reported for E4 (LoRA-full-ablation) is a coarse
- The E2 patch-CE evidence for *localization* is weaker than the
- The E2/E3 top-k-unit patch-CE metric loses absolute-magnitude
- A frozen rank-normalized follow-up resolves that measurement defect in
- The anti-Goodhart reading remains a useful vocabulary, but its strong
- Extension to non-group deployment generators (e.g., semantic
- The highest-priority methodological issue is the label-exposure

6.5 Open confound: labeled orbit coverage vs generator learning

E4's compatibility augmentation generates, for each train pair `(x, y = (x + offset) mod n)`, the pair `((x+k) mod n, (y+k) mod n)` for random `k`. Because `(y+k) = ((x+k) + offset) mod n`, every augmented pair is a *correctly labeled* example at input `(x+k) mod n` — including inputs in the held-out deployment complement. The same holds for E2's `((a+k) mod n, b) ■ (a+b+k) mod n` augmentation. In other words, the intervention arms were trained with direct labeled exposure to the OOD support. The wrong-group Arm C matches augmentation *volume* but places *incorrect* labels on held-out inputs (π`(truth(x))` at input π`(x)` is generally not `truth(`π`(x)))`, so the B-vs-C contrast rules out generic augmentation volume — it does **not** rule out target-support label exposure as the operative mechanism. An earlier revision of Arm C that used correctly-labeled coverage augmentation *did* produce OOD generalization (Appendix A.4, R2), which keeps the coverage explanation live rather than hypothetical.

E4 is therefore consistent with two readings:

generator (cyclic equivariance) and uses it at commitment; the augmentation is how the generator is installed.

OOD orbit with correct labels; nothing transportable is learned beyond fitting the exposed points.

- **Commitment-first (ours).** The model learns a transportable
- **Coverage (deflationary).** Aligned augmentation simply exposes the

The timestamped, explicitly post-hoc E5 addendum to `PLAN.md` (frozen 2026-07-09 21:36 EDT, before any E5 result) preregisters the severe follow-up. Train **without ever presenting labels from the held-out deployment orbit**, and compare five arms:

(e.g., an equivariance-consistency loss `f((x+k) mod n)` vs `(f(x)+k) mod n` evaluated only where both points are in the train support, or where the target is the model's own prediction rather than a ground-truth label);

(the current Arm B, as the coverage-exposed reference);

structure, matched count of held-out-support exposures);

- aligned generator regularization using only train-support pairs
- aligned augmentation that is allowed to enter held-out support
- wrong-generator regularization (matched form of (1), wrong group);
- ordinary coverage-matched augmentation (correct labels, no group
- readout selection (current Arm A).

Evaluate on **new group elements not used by the intervention**, repeat the commitment under task-preserving prompt paraphrases, and use spectral-mass-normalized LoRA causal patching. The harness records support exposure counts and invalidates any G-reg/W-reg cell that receives a held-out truth label.

Kill criteria (pre-registered). The commitment-first interpretation is materially weakened if any of:

deployment-support points (i.e., (1) ≈ (5) while (2) >> (5));

augmentation (2);

- arm (1)'s gains disappear when augmentation cannot label
- coverage-matched augmentation (4) performs on par with aligned
- patch-CE fails to predict transfer to a novel commitment surface.

Until this follow-up runs, the E4 claim should be read as: *train-time aligned intervention recovers external OOD where readout selection does not* — with the mechanism (transportable generator vs labeled coverage) not yet isolated. **E5 status at this revision: the Pythia-70m/n=13/one-seed 20-epoch validation smoke passed its integrity gate.** G-reg recorded zero held-out truth-label exposures, B-ref/Cov's precomputed ledgers matched at 27 held-out events over six unique inputs, novel shifts were disjoint, and the 50% spectral-mass patch tolerance passed. Descriptively, G-reg/Cov/A-ref canonical OOD accuracies were 0.000/0.286/0.000; this undertrained one-seed smoke omits B-ref and W-reg and is explicitly not confirmatory evidence. The generator-vs-coverage mechanism therefore remains pending, and E4's claim boundary is unchanged.

7. Conclusion

Availability is not use; probe AUC is not causal path; portable weakness is not portable simplicity. Structure-in-representation is a starting point, not the discovery. The discovery — the thing you would defend to a skeptical reviewer three moves out — is *commitment-surface survival*: did the causal effect of the identified mechanism survive transport, gauge-fix, and change of commitment on a system you did not build? If yes, the geometry story becomes an aligned-generator special case with clean group theory. If no, the geometry story was a footprint of the prior. Both are useful, but only the first is science.

8. Author's stance

Attribution: human director Jawaun Brown. Code, sweeps, and paper drafts are agent-generated under human direction and review; results are committed with per-cell JSON provenance and pre-registered gates. Prior program overreach (portable scalar concern, weakness-as-universal-law readings) is corrected here in the direction indicated by the anomaly map in `notes/weakness_topology_program_synthesis.md and the pre-registered kills in docs/external_contact_preregistration.md`.

Appendix

A.1 Full derivations for Props 1 and 2

Prop. 1 (non-identification of causal use by probe AUC). Let M_α be the set of admissible hypotheses. Fix a probe $p : X \to [0,1]$ and any admissible baseline f' with a known load-bearing mechanism $M(f')$ such that $CE(f' \mid patch, C) \geq \varepsilon$ when the patch zeroes $M(f')$. We construct f as follows: (i) duplicate f''s decision head onto a fresh subspace S orthogonal to $M(f')$ in feature space; (ii) copy f''s decisions bit-for-bit onto $M(f') \oplus S$; (iii) route the probe input through S while keeping the downstream computation routed through $M(f')$. By construction:

Property (a). $AUC(p \cdot f) = AUC(p \cdot f')$ on any input distribution, because p reads S and S was constructed to match f''s probe response. Property (b). Zero-ablating S changes $p \cdot f$ but not the decision head's output, so $CE(f \mid patch(S), C) = 0$. Property (c). Zero-ablating $M(f')$ changes both the decision output and the probe response, so $CE(f \mid patch(M(f')), C) = CE(f' \mid patch, C) \geq \varepsilon$.

This exhibits two admissible hypotheses with identical probe AUC and causal-effect gap ε under the respective identified-mechanism patches. To realize an arbitrary intermediate gap $\delta \in [0, \varepsilon]$, interpolate the *routing*, not the functions: construct f_λ that routes a fraction λ of the decision head's input mass through $M(f')$ and $1 - \lambda$ through the (causally inert, probe-visible) copy on S, with the head re-normalized so the input–output behavior — hence admissibility and probe response — is unchanged. $AUC(p \cdot f_\lambda)$ is constant in λ while $CE(f_\lambda \mid patch(S), C)$ moves continuously from 0 (at $\lambda = 1$) toward ε (at $\lambda = 0$), so every gap $\delta \in [0, \varepsilon]$ is attained at equal AUC.

Hence no function of probe AUC alone identifies, or lower-bounds, $CE(\cdot \mid patch, C)$ on M_α. We emphasize the scope: this is a *non-identification* theorem over the admissible class. It does **not** assert probabilistic independence of AUC and CE — that would require a distribution over M_α, and under particular training distributions the two can correlate (E3 measures such a correlation in the aligned regime). What fails is the inference from probe evidence to causal use for any *individual* hypothesis. ■

Prop. 2 (weakness is diagnostic exactly when aligned). *Deployment model.* Both groups finite. A deployment slice is generated by drawing $g \sim Uniform(G_{dep})$ (the prior) and evaluating the candidate on the g-translated support; the candidate scores 1 on the slice iff it is compatible with g (the orbit-likelihood), and concern is uniform. Under this model the expected deployment accuracy of an admissible f is

$$E_g[acc(f)] = W_{G_{dep}}(f) / |G_{dep}|,$$

i.e. expected accuracy is *exactly* normalized deployment-restricted weakness.

(i) Aligned case. If the selector scores candidates by $W_{G_{dep}}$ — either because $G_{probe} = G_{dep}$, or because weakness is explicitly restricted (or concern-weighted) to G_{dep} — then $\arg\max_f W_{G_{dep}}(f)$ maximizes $E_g[acc(f)]$ by the identity above. Since the score is a deterministic function of f and the objective is its own posterior expectation under the stated prior and likelihood, the argmax selector is Bayes-optimal among selectors that depend only on f. (Bayes-optimality here is *relative to this explicit prior and likelihood*; no claim is made under other priors.)

(ii) Superset counterexample. $G_{probe} \supset G_{dep}$ does not suffice. Let $G_{dep} = \{e, g\}$ and $G_{probe} = \{e, g, h1, h2, h3\}$. Take f compatible with $\{e, h1, h2, h3\}$ (so $W_{G_{probe}}(f) = 4$, $W_{G_{dep}}(f) = 1$) and f' compatible with $\{e, g\}$ (so $W_{G_{probe}}(f') = 2$, $W_{G_{dep}}(f') = 2$). Then $W_{G_{probe}}(f) > W_{G_{probe}}(f')$ while $E_g[acc(f)] = 1/2 < 1 = E_g[acc(f')]$: the superset-probe ranking is inverted on deployment. Optimality from a superset probe group is recovered only under an ordering-preservation assumption on the candidate set: $W_{G_{probe}}(f) > W_{G_{probe}}(f') \Rightarrow W_{G_{dep}}(f) \geq W_{G_{dep}}(f')$.

(iii) Non-aligned case. Suppose $G_{dep} \not\subset G_{probe}$. Let $g' \in G_{dep} \setminus G_{probe}$. Then $W_{G_{probe}}(f)$ does not measure compatibility with g'. Construct f, f' admissible such that $W_{G_{probe}}(f) > W_{G_{probe}}(f')$ but f' is compatible with g' and f is not; then f' has strictly higher expected deployment accuracy on the g'-generated slice while f has higher $W_{G_{probe}}$ — a negative association between selector score and OOD accuracy on that candidate pair. Hence $W_{G_{probe}}$ is a footprint whose sign of association with deployment accuracy is generator-dependent. ■

Corollary (order-equivalence for concern-weighted extension mass). Let `C : U → R_≥0` be a concern measure on the deployment slice `U`, `C_star` the true consequence measure, and `F ⊂ M_α` a finite candidate set. *Sufficiency:* if `C = a·C_star` for some `a > 0`, then `W_C = a·W_{C_star}` and `argmax_F W_C = argmax_F W_{C_star}`, so the selector is `C_star`-optimal on `F`. *Exact condition:* the selector `argmax_F W_C` picks a `C_star`-optimal candidate for every truth assignment iff `C` is order-equivalent to `C_star` on `F`:

`W_C(f, U) > W_C(f', U) ⇔ W_{C_star}(f, U) > W_{C_star}(f', U)` for all `f, f' ∈ F`.

Equality up to positive scale is *not necessary*: over a finite `F`, distinct weightings that induce the same ranking of candidates are equally optimal, so the earlier "iff `C = C_star` up to scaling" phrasing was too strong and is retracted here in favor of the order-equivalence condition. *Misspecification:* if `C` is an exchangeable random assignment with the same coordinate marginal as `C_star`, independent of candidate coverage, then for every fixed candidate `E[W_C(f, U)] = E[C] · |{x ∈ U : f(x) = truth(x)}|` — proportional to unweighted extension mass. Fixed-cardinality assignment is exchangeable but not iid across positions; equal marginals are sufficient for this identity. However, `E[argmax_f W_C(f,U)]` is not determined by `argmax_f E[W_C(f,U)]`: finite candidate sets, ties, and winner selection make argmax nonlinear. The E1 follow-up measures this distinction directly. Across 2,048 independent reassignments with all 96 candidate/deployment structures frozen, the selected-performance gap has mean −0.0589 and the observed −0.054 has lower-tail probability 0.620. Thus E1 supports the score-level identity but falsifies the stronger, previously implied selector-equality reading for this finite candidate design. A strictly adversarial `C` (anti-correlated with `C_star`) can still invert the relation, but the observed assignment provides no evidence for that mechanism. ■

A.2 Per-cell tables

The PDF build renders complete per-cell tables directly from the committed public-safe E1 and E2/E3 JSON summaries and the compact committed E4 appendix artifact. The E2 and E3 analyses share one 216-cell population. E4 exports all 108 cells while intentionally omitting raw function tables, train/OOD input lists, and parameter metadata; no requested appendix metric is unavailable.

A.2.1 E1. All 96 concern-weighted selector cells from `experiments/commitment_surface/results/e1_concern_weighted.json`.

n	Seed	Unweighted	Well-spec	Misspec	Loss	Truth
7	0	0.591	0.848	0.515	0.197	1.000
7	1	0.621	0.864	0.561	0.364	1.000
7	2	0.879	0.879	0.545	0.409	1.000
7	3	0.470	0.864	0.409	0.045	1.000
7	4	0.591	0.848	0.561	0.591	1.000
7	5	0.606	0.879	0.606	0.076	1.000
7	6	0.621	0.848	0.455	0.061	1.000
7	7	0.318	0.848	0.318	0.379	1.000
7	8	0.591	0.864	0.576	0.424	1.000
7	9	0.864	0.864	0.439	0.500	1.000
7	10	0.591	0.818	0.727	0.500	1.000
7	11	0.606	0.879	0.682	0.076	1.000
7	12	0.742	0.848	0.455	0.364	1.000
7	13	0.606	0.864	0.697	0.364	1.000
7	14	0.591	0.833	0.288	0.394	1.000
7	15	0.591	0.864	0.303	0.227	1.000
7	16	0.621	0.879	0.712	0.288	1.000
7	17	0.742	0.864	0.394	0.227	1.000
7	18	0.879	0.879	0.727	0.545	1.000
7	19	0.439	0.848	0.682	0.227	1.000
7	20	0.333	0.848	0.333	0.545	1.000
7	21	0.455	0.848	0.424	0.227	1.000
7	22	0.879	0.879	0.879	0.212	1.000
7	23	0.758	0.833	0.576	0.212	1.000
7	24	0.606	0.864	0.712	0.212	1.000
7	25	0.712	0.848	0.712	0.045	1.000
7	26	0.742	0.864	0.591	0.242	1.000
7	27	0.606	0.833	0.530	0.485	1.000
7	28	0.182	0.833	0.561	0.561	1.000
7	29	0.727	0.848	0.727	0.182	1.000

n	Seed	Unweighted	Well-spec	Misspec	Loss	Truth
7	30	0.742	0.879	0.409	0.227	1.000
7	31	0.591	0.848	0.409	0.273	1.000
11	0	0.547	0.796	0.619	0.238	1.000
11	1	0.564	0.829	0.564	0.271	1.000
11	2	0.862	0.862	0.862	0.271	1.000
11	3	0.646	0.779	0.497	0.309	1.000
11	4	0.398	0.812	0.381	0.453	1.000
11	5	0.790	0.823	0.779	0.177	1.000
11	6	0.503	0.773	0.326	0.370	1.000
11	7	0.459	0.823	0.381	0.359	1.000
11	8	0.448	0.840	0.779	0.381	1.000
11	9	0.343	0.773	0.630	0.448	1.000
11	10	0.746	0.818	0.619	0.448	1.000
11	11	0.547	0.807	0.497	0.315	1.000
11	12	0.547	0.818	0.370	0.293	1.000
11	13	0.602	0.818	0.387	0.232	1.000
11	14	0.796	0.829	0.381	0.304	1.000
11	15	0.796	0.796	0.718	0.293	1.000
11	16	0.602	0.785	0.602	0.409	1.000
11	17	0.442	0.823	0.453	0.199	1.000
11	18	0.696	0.840	0.470	0.155	1.000
11	19	0.602	0.790	0.271	0.343	1.000
11	20	0.547	0.829	0.431	0.420	1.000
11	21	0.453	0.829	0.326	0.403	1.000
11	22	0.840	0.840	0.834	0.182	1.000
11	23	0.287	0.807	0.271	0.354	1.000
11	24	0.381	0.762	0.680	0.260	1.000
11	25	0.696	0.823	0.442	0.166	1.000
11	26	0.608	0.773	0.392	0.304	1.000
11	27	0.564	0.845	0.530	0.514	1.000
11	28	0.442	0.812	0.724	0.315	1.000
11	29	0.602	0.829	0.376	0.320	1.000
11	30	0.597	0.829	0.376	0.199	1.000
11	31	0.387	0.812	0.238	0.414	1.000
13	0	0.409	0.795	0.735	0.201	1.000
13	1	0.493	0.752	0.493	0.292	1.000
13	2	0.406	0.752	0.735	0.326	1.000
13	3	0.436	0.789	0.352	0.144	1.000
13	4	0.440	0.765	0.426	0.430	1.000
13	5	0.678	0.779	0.634	0.302	1.000
13	6	0.403	0.785	0.295	0.174	1.000
13	7	0.477	0.738	0.477	0.349	1.000
13	8	0.752	0.755	0.406	0.470	1.000
13	9	0.648	0.748	0.480	0.302	1.000
13	10	0.537	0.805	0.218	0.268	1.000
13	11	0.346	0.792	0.792	0.262	1.000
13	12	0.312	0.758	0.362	0.326	1.000
13	13	0.409	0.822	0.376	0.268	1.000
13	14	0.560	0.795	0.795	0.416	1.000
13	15	0.678	0.789	0.463	0.292	1.000
13	16	0.470	0.762	0.436	0.493	1.000
13	17	0.688	0.779	0.349	0.349	1.000
13	18	0.463	0.779	0.255	0.205	1.000
13	19	0.738	0.768	0.661	0.295	1.000
13	20	0.792	0.792	0.735	0.315	1.000
13	21	0.450	0.779	0.450	0.272	1.000
13	22	0.356	0.775	0.671	0.302	1.000
13	23	0.685	0.728	0.594	0.406	1.000
13	24	0.386	0.792	0.346	0.265	1.000
13	25	0.725	0.762	0.725	0.262	1.000
13	26	0.735	0.762	0.557	0.443	1.000

n	Seed	Unweighted	Well-spec	Misspec	Loss	Truth
13	27	0.285	0.738	0.315	0.258	1.000
13	28	0.473	0.762	0.436	0.309	1.000
13	29	0.386	0.799	0.406	0.285	1.000
13	30	0.466	0.775	0.416	0.342	1.000
13	31	0.419	0.772	0.419	0.295	1.000

A.2.2 E2/E3. All 216 trained neural cells. E2 arm comparisons and E3 correlations use this same committed cell population from `experiments/commitment_surface/results/e2_e3_neural.json`.

Arm	n	Seed	Train frac	OOD	Patch-CE	Wrong patch	Weakness
A	7	20261080	0.40	0.000	-0.583	-0.649	0.143
B	7	20261080	0.40	1.000	0.051	0.002	1.000
C	7	20261080	0.40	0.172	0.000	0.000	0.143
D	7	20261080	0.40	0.000	-0.583	-0.649	0.143
A	7	20261181	0.40	0.000	0.000	0.000	0.143
B	7	20261181	0.40	1.000	0.027	0.002	1.000
C	7	20261181	0.40	0.138	0.000	0.000	0.143
D	7	20261181	0.40	0.000	0.000	0.000	0.143
A	7	20261282	0.40	0.000	0.000	0.000	0.143
B	7	20261282	0.40	1.000	0.015	0.001	1.000
C	7	20261282	0.40	0.138	0.000	0.000	0.143
D	7	20261282	0.40	0.000	0.000	0.000	0.143
A	7	20261383	0.40	0.000	-0.138	-0.338	0.143
B	7	20261383	0.40	1.000	0.029	0.005	1.000
C	7	20261383	0.40	0.310	0.000	0.000	0.143
D	7	20261383	0.40	0.000	-0.138	-0.338	0.143
A	7	20261484	0.40	0.000	-0.212	-0.438	0.143
B	7	20261484	0.40	0.759	-0.165	-0.096	0.143
C	7	20261484	0.40	0.103	0.000	0.000	0.143
D	7	20261484	0.40	0.000	-0.212	-0.438	0.143
A	7	20261585	0.40	0.000	-0.323	-0.274	0.143
B	7	20261585	0.40	1.000	0.006	0.002	1.000
C	7	20261585	0.40	0.310	0.000	0.000	0.143
D	7	20261585	0.40	0.000	-0.323	-0.274	0.143
A	7	20261185	0.55	0.000	-1.453	-0.659	0.143
B	7	20261185	0.55	1.000	0.028	0.002	1.000
C	7	20261185	0.55	0.318	0.000	0.000	0.143
D	7	20261185	0.55	0.000	-1.453	-0.659	0.143
A	7	20261286	0.55	0.000	0.191	-0.254	0.143
B	7	20261286	0.55	1.000	0.022	0.001	1.000
C	7	20261286	0.55	0.273	-0.006	0.013	0.143
D	7	20261286	0.55	0.000	0.191	-0.254	0.143
A	7	20261387	0.55	0.000	-0.161	-0.313	0.143
B	7	20261387	0.55	1.000	0.026	0.007	1.000
C	7	20261387	0.55	0.182	0.013	0.025	0.143
D	7	20261387	0.55	0.000	-0.161	-0.313	0.143
A	7	20261488	0.55	0.000	-0.859	-0.469	0.143
B	7	20261488	0.55	1.000	0.029	0.001	1.000
C	7	20261488	0.55	0.227	0.022	0.002	0.143
D	7	20261488	0.55	0.000	-0.859	-0.469	0.143
A	7	20261589	0.55	0.000	0.000	0.000	0.143
B	7	20261589	0.55	1.000	0.032	0.001	1.000
C	7	20261589	0.55	0.182	0.000	0.000	0.143
D	7	20261589	0.55	0.000	0.000	0.000	0.143
A	7	20261690	0.55	0.000	-0.160	-0.717	0.143
B	7	20261690	0.55	1.000	0.004	0.001	1.000
C	7	20261690	0.55	0.318	-0.001	0.001	0.143
D	7	20261690	0.55	0.000	-0.160	-0.717	0.143
A	7	20261290	0.70	0.000	-1.012	-1.279	0.143
B	7	20261290	0.70	1.000	0.023	0.003	1.000
C	7	20261290	0.70	0.067	-0.004	-0.004	0.143
D	7	20261290	0.70	0.000	-1.012	-1.279	0.143
A	7	20261391	0.70	0.000	-0.481	-1.114	0.143
B	7	20261391	0.70	1.000	0.014	0.008	1.000
C	7	20261391	0.70	0.200	-0.006	-0.012	0.143
D	7	20261391	0.70	0.000	-0.481	-1.114	0.143
A	7	20261492	0.70	0.000	-1.182	-1.427	0.143
B	7	20261492	0.70	1.000	0.014	0.001	1.000

Arm	n	Seed	Train frac	OOD	Patch-CE	Wrong patch	Weakness
C	7	20261492	0.70	0.200	-0.007	0.008	0.143
D	7	20261492	0.70	0.000	-1.182	-1.427	0.143
A	7	20261593	0.70	0.000	-0.421	-0.857	0.143
B	7	20261593	0.70	1.000	0.028	0.002	1.000
C	7	20261593	0.70	0.067	-0.008	-0.006	0.143
D	7	20261593	0.70	0.000	-0.421	-0.857	0.143
A	7	20261694	0.70	0.000	0.058	-1.453	0.143
B	7	20261694	0.70	1.000	0.016	0.001	1.000
C	7	20261694	0.70	0.200	-0.001	-0.019	0.143
D	7	20261694	0.70	0.000	0.058	-1.453	0.143
A	7	20261795	0.70	0.000	-0.580	-0.541	0.143
B	7	20261795	0.70	1.000	0.070	0.002	1.000
C	7	20261795	0.70	0.200	-0.010	0.016	0.143
D	7	20261795	0.70	0.000	-0.580	-0.541	0.143
A	11	20261132	0.40	0.000	-0.906	-0.617	0.091
B	11	20261132	0.40	1.000	0.015	0.008	1.000
C	11	20261132	0.40	0.164	-0.003	0.014	0.091
D	11	20261132	0.40	0.000	-0.906	-0.617	0.091
A	11	20261233	0.40	0.000	-0.748	-0.707	0.091
B	11	20261233	0.40	1.000	0.048	0.003	1.000
C	11	20261233	0.40	0.151	0.000	0.007	0.091
D	11	20261233	0.40	0.000	-0.748	-0.707	0.091
A	11	20261334	0.40	0.000	-0.328	-0.346	0.091
B	11	20261334	0.40	1.000	0.013	0.002	1.000
C	11	20261334	0.40	0.151	0.000	0.000	0.091
D	11	20261334	0.40	0.000	-0.328	-0.346	0.091
A	11	20261435	0.40	0.000	-0.799	-1.100	0.091
B	11	20261435	0.40	1.000	0.029	0.008	1.000
C	11	20261435	0.40	0.151	0.008	0.011	0.091
D	11	20261435	0.40	0.000	-0.799	-1.100	0.091
A	11	20261536	0.40	0.000	-0.160	-0.941	0.091
B	11	20261536	0.40	1.000	0.016	0.005	1.000
C	11	20261536	0.40	0.164	0.006	0.010	0.091
D	11	20261536	0.40	0.000	-0.160	-0.941	0.091
A	11	20261637	0.40	0.000	-0.534	-0.985	0.091
B	11	20261637	0.40	1.000	0.005	0.004	1.000
C	11	20261637	0.40	0.151	0.008	0.029	0.091
D	11	20261637	0.40	0.000	-0.534	-0.985	0.091
A	11	20261237	0.55	0.000	-0.961	-0.678	0.091
B	11	20261237	0.55	1.000	0.058	0.003	1.000
C	11	20261237	0.55	0.204	-0.001	0.007	0.091
D	11	20261237	0.55	0.000	-0.961	-0.678	0.091
A	11	20261338	0.55	0.000	-0.004	-1.338	0.091
B	11	20261338	0.55	1.000	0.033	0.005	1.000
C	11	20261338	0.55	0.074	0.001	0.003	0.091
D	11	20261338	0.55	0.000	-0.004	-1.338	0.091
A	11	20261439	0.55	0.000	-0.307	-0.327	0.091
B	11	20261439	0.55	1.000	0.006	0.001	1.000
C	11	20261439	0.55	0.148	-0.004	0.022	0.091
D	11	20261439	0.55	0.000	-0.307	-0.327	0.091
A	11	20261540	0.55	0.000	-0.881	-0.980	0.091
B	11	20261540	0.55	1.000	0.011	0.006	1.000
C	11	20261540	0.55	0.130	0.015	0.027	0.091
D	11	20261540	0.55	0.000	-0.881	-0.980	0.091
A	11	20261641	0.55	0.000	-0.840	-0.920	0.091
B	11	20261641	0.55	1.000	0.074	0.003	1.000
C	11	20261641	0.55	0.204	0.009	-0.020	0.091
D	11	20261641	0.55	0.000	-0.840	-0.920	0.091
A	11	20261742	0.55	0.000	-0.844	-0.813	0.091
B	11	20261742	0.55	1.000	0.039	0.007	1.000
C	11	20261742	0.55	0.148	-0.006	0.013	0.091

Arm	n	Seed	Train frac	OOD	Patch-CE	Wrong patch	Weakness
D	11	20261742	0.55	0.000	-0.844	-0.813	0.091
A	11	20261342	0.70	0.000	-0.623	-0.854	0.091
B	11	20261342	0.70	1.000	0.022	0.010	1.000
C	11	20261342	0.70	0.167	0.003	0.006	0.091
D	11	20261342	0.70	0.000	-0.623	-0.854	0.091
A	11	20261443	0.70	0.000	-0.519	-1.052	0.091
B	11	20261443	0.70	1.000	0.008	0.006	1.000
C	11	20261443	0.70	0.083	0.023	0.017	0.091
D	11	20261443	0.70	0.000	-0.519	-1.052	0.091
A	11	20261544	0.70	0.000	-0.346	-0.801	0.091
B	11	20261544	0.70	1.000	0.019	0.002	1.000
C	11	20261544	0.70	0.139	0.040	0.042	0.091
D	11	20261544	0.70	0.000	-0.346	-0.801	0.091
A	11	20261645	0.70	0.000	-0.401	-0.687	0.091
B	11	20261645	0.70	1.000	0.137	0.003	1.000
C	11	20261645	0.70	0.250	0.029	0.028	0.091
D	11	20261645	0.70	0.000	-0.401	-0.687	0.091
A	11	20261746	0.70	0.000	-0.535	-0.807	0.091
B	11	20261746	0.70	1.000	0.042	0.002	1.000
C	11	20261746	0.70	0.250	0.014	0.002	0.091
D	11	20261746	0.70	0.000	-0.535	-0.807	0.091
A	11	20261847	0.70	0.000	-0.444	-0.903	0.091
B	11	20261847	0.70	1.000	0.004	0.002	1.000
C	11	20261847	0.70	0.056	-0.015	0.034	0.091
D	11	20261847	0.70	0.000	-0.444	-0.903	0.091
A	13	20261158	0.40	0.000	-0.438	-0.527	0.077
B	13	20261158	0.40	0.871	-0.055	-0.030	0.077
C	13	20261158	0.40	0.119	-0.000	0.024	0.077
D	13	20261158	0.40	0.000	-0.438	-0.527	0.077
A	13	20261259	0.40	0.000	-0.733	-0.365	0.077
B	13	20261259	0.40	1.000	0.016	0.010	1.000
C	13	20261259	0.40	0.149	0.000	-0.002	0.077
D	13	20261259	0.40	0.000	-0.733	-0.365	0.077
A	13	20261360	0.40	0.000	-0.187	-0.586	0.077
B	13	20261360	0.40	1.000	0.043	0.009	1.000
C	13	20261360	0.40	0.129	0.007	0.009	0.077
D	13	20261360	0.40	0.000	-0.187	-0.586	0.077
A	13	20261461	0.40	0.000	-1.106	-0.529	0.077
B	13	20261461	0.40	1.000	0.026	0.006	1.000
C	13	20261461	0.40	0.149	0.000	0.000	0.077
D	13	20261461	0.40	0.000	-1.106	-0.529	0.077
A	13	20261562	0.40	0.000	-0.156	-0.597	0.077
B	13	20261562	0.40	1.000	0.014	0.004	1.000
C	13	20261562	0.40	0.109	-0.004	-0.000	0.077
D	13	20261562	0.40	0.000	-0.156	-0.597	0.077
A	13	20261663	0.40	0.000	-1.055	-0.989	0.077
B	13	20261663	0.40	1.000	0.056	0.003	1.000
C	13	20261663	0.40	0.149	0.002	0.025	0.077
D	13	20261663	0.40	0.000	-1.055	-0.989	0.077
A	13	20261263	0.55	0.000	-0.305	-0.734	0.077
B	13	20261263	0.55	1.000	0.059	0.007	1.000
C	13	20261263	0.55	0.158	-0.007	0.011	0.077
D	13	20261263	0.55	0.000	-0.305	-0.734	0.077
A	13	20261364	0.55	0.000	-1.158	-0.590	0.077
B	13	20261364	0.55	1.000	0.014	0.003	1.000
C	13	20261364	0.55	0.118	0.018	0.016	0.077
D	13	20261364	0.55	0.000	-1.158	-0.590	0.077
A	13	20261465	0.55	0.000	-0.333	-1.076	0.077
B	13	20261465	0.55	1.000	0.029	0.010	1.000
C	13	20261465	0.55	0.197	-0.009	0.016	0.077
D	13	20261465	0.55	0.000	-0.333	-1.076	0.077

Arm	n	Seed	Train frac	OOD	Patch-CE	Wrong patch	Weakness
A	13	20261566	0.55	0.000	-0.777	-1.066	0.077
B	13	20261566	0.55	1.000	0.021	0.008	1.000
C	13	20261566	0.55	0.145	0.004	0.018	0.077
D	13	20261566	0.55	0.000	-0.777	-1.066	0.077
A	13	20261667	0.55	0.000	-0.382	-0.717	0.077
B	13	20261667	0.55	1.000	0.016	0.002	1.000
C	13	20261667	0.55	0.224	0.000	0.000	0.077
D	13	20261667	0.55	0.000	-0.382	-0.717	0.077
A	13	20261768	0.55	0.000	-0.865	-1.136	0.077
B	13	20261768	0.55	1.000	0.013	0.005	1.000
C	13	20261768	0.55	0.171	-0.001	0.006	0.077
D	13	20261768	0.55	0.000	-0.865	-1.136	0.077
A	13	20261368	0.70	0.000	-0.327	-0.724	0.077
B	13	20261368	0.70	1.000	0.058	0.003	1.000
C	13	20261368	0.70	0.157	0.005	0.036	0.077
D	13	20261368	0.70	0.000	-0.327	-0.724	0.077
A	13	20261469	0.70	0.000	-0.257	-1.076	0.077
B	13	20261469	0.70	1.000	0.020	0.003	1.000
C	13	20261469	0.70	0.216	0.013	0.004	0.077
D	13	20261469	0.70	0.000	-0.257	-1.076	0.077
A	13	20261570	0.70	0.000	-0.316	-0.734	0.077
B	13	20261570	0.70	1.000	0.006	0.004	1.000
C	13	20261570	0.70	0.118	0.012	0.015	0.077
D	13	20261570	0.70	0.000	-0.316	-0.734	0.077
A	13	20261671	0.70	0.000	-0.202	-0.732	0.077
B	13	20261671	0.70	1.000	0.031	0.001	1.000
C	13	20261671	0.70	0.118	0.007	0.177	0.077
D	13	20261671	0.70	0.000	-0.202	-0.732	0.077
A	13	20261772	0.70	0.000	-0.463	-0.725	0.077
B	13	20261772	0.70	1.000	0.031	0.005	1.000
C	13	20261772	0.70	0.098	0.015	0.056	0.077
D	13	20261772	0.70	0.000	-0.463	-0.725	0.077
A	13	20261873	0.70	0.000	-0.694	-0.742	0.077
B	13	20261873	0.70	1.000	0.054	0.004	1.000
C	13	20261873	0.70	0.118	-0.019	0.089	0.077
D	13	20261873	0.70	0.000	-0.694	-0.742	0.077

A.2.3 E4. All 108 Pythia LoRA cells from `experiments/commitment_surface/results/e4_pythia_lora_v2_appendix.json`. No requested appendix fields are unavailable.

Size	Arm	n	Seed	OOD	Ablated OOD	Patch-CE	Weakness	Wrong group
70m	A	13	20260709	0.000	0.000	-2.678	0.077	0.077
70m	B	13	20260709	0.714	0.000	4.703	0.077	0.077
70m	C	13	20260709	0.143	0.000	4.140	0.077	0.077
70m	D	13	20260709	0.000	0.000	-2.678	0.077	0.077
70m	A	13	20260809	0.000	0.143	-4.267	0.077	0.077
70m	B	13	20260809	0.857	0.143	4.995	0.077	0.077
70m	C	13	20260809	0.000	0.143	1.327	0.077	0.077
70m	D	13	20260809	0.000	0.143	-4.267	0.077	0.077
70m	A	13	20260909	0.000	0.000	-0.952	0.077	0.077
70m	B	13	20260909	0.857	0.000	5.097	0.077	0.077
70m	C	13	20260909	0.000	0.000	2.638	0.077	0.077
70m	D	13	20260909	0.000	0.000	-0.952	0.077	0.077
70m	A	17	20260709	0.000	0.000	-2.509	0.059	0.059
70m	B	17	20260709	0.667	0.000	4.223	0.059	0.059
70m	C	17	20260709	0.000	0.000	1.092	0.059	0.059
70m	D	17	20260709	0.000	0.000	-2.509	0.059	0.059
70m	A	17	20260809	0.000	0.000	-3.455	0.059	0.059
70m	B	17	20260809	0.889	0.000	5.711	0.059	0.059
70m	C	17	20260809	0.222	0.000	3.404	0.059	0.059
70m	D	17	20260809	0.000	0.000	-3.455	0.059	0.059
70m	A	17	20260909	0.000	0.111	-2.797	0.059	0.059
70m	B	17	20260909	0.889	0.111	5.092	0.059	0.059
70m	C	17	20260909	0.111	0.111	1.837	0.059	0.059
70m	D	17	20260909	0.000	0.111	-2.797	0.059	0.059
70m	A	23	20260709	0.000	0.091	-1.068	0.043	0.043
70m	B	23	20260709	0.909	0.091	5.986	0.043	0.043
70m	C	23	20260709	0.000	0.091	0.628	0.043	0.043
70m	D	23	20260709	0.000	0.091	-1.068	0.043	0.043
70m	A	23	20260809	0.000	0.000	-0.342	0.043	0.043
70m	B	23	20260809	1.000	0.000	6.474	1.000	0.043
70m	C	23	20260809	0.091	0.000	1.870	0.043	0.043
70m	D	23	20260809	0.000	0.000	-0.342	0.043	0.043
70m	A	23	20260909	0.000	0.000	0.646	0.043	0.043
70m	B	23	20260909	0.818	0.000	6.017	0.043	0.043
70m	C	23	20260909	0.091	0.000	2.270	0.043	0.043
70m	D	23	20260909	0.000	0.000	0.646	0.043	0.043
160m	A	13	20260709	0.000	0.000	-7.546	0.077	0.077
160m	B	13	20260709	0.714	0.000	3.415	0.077	0.077
160m	C	13	20260709	0.143	0.000	2.755	0.077	0.077
160m	D	13	20260709	0.000	0.000	-7.546	0.077	0.077
160m	A	13	20260809	0.143	0.000	-1.909	0.077	0.077
160m	B	13	20260809	1.000	0.000	5.396	1.000	0.077
160m	C	13	20260809	0.000	0.000	1.645	0.077	0.077
160m	D	13	20260809	0.143	0.000	-1.909	0.077	0.077
160m	A	13	20260909	0.000	0.000	-1.527	0.077	0.077
160m	B	13	20260909	0.857	0.000	5.047	0.077	0.077
160m	C	13	20260909	0.143	0.000	2.816	0.077	0.077
160m	D	13	20260909	0.000	0.000	-1.527	0.077	0.077
160m	A	17	20260709	0.556	0.000	4.414	0.059	0.059
160m	B	17	20260709	0.889	0.000	4.769	0.059	0.059
160m	C	17	20260709	0.000	0.000	1.646	0.059	0.059
160m	D	17	20260709	0.556	0.000	4.414	0.059	0.059
160m	A	17	20260809	0.000	0.000	-2.674	0.059	0.059
160m	B	17	20260809	0.889	0.000	5.465	0.059	0.059
160m	C	17	20260809	0.111	0.000	1.369	0.059	0.059
160m	D	17	20260809	0.000	0.000	-2.674	0.059	0.059
160m	A	17	20260909	0.000	0.111	-4.009	0.059	0.059

Size	Arm	n	Seed	OOD	Ablated OOD	Patch-CE	Weakness	Wrong group
160m	B	17	20260909	0.889	0.111	4.308	0.059	0.059
160m	C	17	20260909	0.111	0.111	1.140	0.059	0.059
160m	D	17	20260909	0.000	0.111	-4.009	0.059	0.059
160m	A	23	20260709	0.000	0.000	-1.967	0.043	0.043
160m	B	23	20260709	1.000	0.000	5.735	1.000	0.043
160m	C	23	20260709	0.000	0.000	1.956	0.043	0.043
160m	D	23	20260709	0.000	0.000	-1.967	0.043	0.043
160m	A	23	20260809	0.000	0.000	0.199	0.043	0.043
160m	B	23	20260809	1.000	0.000	5.849	1.000	0.043
160m	C	23	20260809	0.091	0.000	1.057	0.043	0.043
160m	D	23	20260809	0.000	0.000	0.199	0.043	0.043
160m	A	23	20260909	0.000	0.000	-1.036	0.043	0.043
160m	B	23	20260909	0.818	0.000	5.527	0.043	0.043
160m	C	23	20260909	0.091	0.000	1.714	0.043	0.043
160m	D	23	20260909	0.000	0.000	-1.036	0.043	0.043
410m	A	13	20260709	0.000	0.000	-0.434	0.077	0.077
410m	B	13	20260709	0.714	0.000	3.171	0.077	0.077
410m	C	13	20260709	0.143	0.000	1.277	0.077	0.077
410m	D	13	20260709	0.000	0.000	-0.434	0.077	0.077
410m	A	13	20260809	0.143	0.000	0.389	0.077	0.077
410m	B	13	20260809	1.000	0.000	4.085	1.000	0.077
410m	C	13	20260809	0.000	0.000	-0.108	0.077	0.077
410m	D	13	20260809	0.143	0.000	0.389	0.077	0.077
410m	A	13	20260909	0.143	0.000	2.405	0.077	0.077
410m	B	13	20260909	0.857	0.000	3.931	0.077	0.077
410m	C	13	20260909	0.143	0.000	0.533	0.077	0.077
410m	D	13	20260909	0.143	0.000	2.405	0.077	0.077
410m	A	17	20260709	1.000	0.222	4.142	1.000	0.059
410m	B	17	20260709	1.000	0.222	4.148	1.000	0.059
410m	C	17	20260709	0.000	0.222	0.598	0.059	0.059
410m	D	17	20260709	1.000	0.222	4.142	1.000	0.059
410m	A	17	20260809	0.000	0.000	1.016	0.059	0.059
410m	B	17	20260809	0.889	0.000	4.384	0.059	0.059
410m	C	17	20260809	0.111	0.000	1.102	0.059	0.059
410m	D	17	20260809	0.000	0.000	1.016	0.059	0.059
410m	A	17	20260909	0.333	0.000	1.286	0.059	0.059
410m	B	17	20260909	0.889	0.000	4.136	0.059	0.059
410m	C	17	20260909	0.000	0.000	0.110	0.059	0.059
410m	D	17	20260909	0.333	0.000	1.286	0.059	0.059
410m	A	23	20260709	0.000	0.000	-0.022	0.043	0.043
410m	B	23	20260709	0.909	0.000	4.368	0.043	0.043
410m	C	23	20260709	0.000	0.000	-0.190	0.043	0.043
410m	D	23	20260709	0.000	0.000	-0.022	0.043	0.043
410m	A	23	20260809	0.455	0.000	2.417	0.043	0.043
410m	B	23	20260809	1.000	0.000	4.661	1.000	0.043
410m	C	23	20260809	0.091	0.000	0.441	0.043	0.043
410m	D	23	20260809	0.455	0.000	2.417	0.043	0.043
410m	A	23	20260909	0.273	0.000	2.241	0.043	0.043
410m	B	23	20260909	0.909	0.000	4.576	0.043	0.043
410m	C	23	20260909	0.091	0.000	-0.056	0.043	0.043
410m	D	23	20260909	0.273	0.000	2.241	0.043	0.043

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A.4 Reviewer response (pre-empt)

We anticipate three lines of criticism and answer here.

R1. "Your cyclic modular addition is a synthetic world; the E4 external contact is still on modular addition, so this isn't external."

Response. The E4 external is *not* another synthetic world. It uses a public open-weights model family (Pythia 70m/160m/410m) trained on The Pile, and asks whether a LoRA fine-tune with cyclic-orbit augmentation produces load-bearing OOD generalization. The old-frame reading of our own prior program predicted a yes on weakness readout; our P1 hard-kill and E4 Arm A both said no. The commitment-first reframe predicted that Arm B (train-time compat intervention) would recover, and E4 does

— at n=13 in smoke, at scale in the full grid. This is exactly the kind of *pre-registered directional prediction on a system the lab did not build* the prior program's critique identified as missing. It is not the last word on external contact — we say so in Section 6.4 — but it is a genuine one.

R2. "Patch-CE is confounded with total LoRA effect; you're just measuring that the model uses fine-tuning."

Response. The wrong-group Arm C is the anti-cheat. Arm C has the same *augmentation volume* as Arm B — same number of extra training pairs, same LoRA capacity, same optimizer trajectory — but the augmented pair `(x, y = truth(x))` is transformed by a random non-cyclic permutation π to `(π(x), π(y))`, teaching the model the wrong equivariance `f(π(x)) = π(f(x))` instead of the cyclic action `f(x + k) = f(x) + k`. On any input that also appears in the base training set with its true label, Arm C's augmentation is *inconsistent* with the cyclic rule, so the model cannot fit both without choosing. If patch-CE were only measuring "the LoRA update is used", Arms B and C would have similar patch-CE and similar OOD. Empirically the anti-cheat holds: LoRA is only load-bearing when the augmentation *group* matches the deployment generator, not merely when augmentation *volume* is present. However, Arm C controls volume, **not label exposure**: Arm B's cyclic augmentation places correctly labeled examples on the held-out deployment support, while Arm C places incorrectly labeled ones there. An earlier revision of Arm C used the labeled coverage augmentation `(π(x), b) → truth(π(x), b)`, which just extends training coverage with correct labels and — as expected — also produced OOD generalization. That result answers a different question ("does adding correct-labeled coverage help?" — yes) and, importantly, keeps the coverage explanation of Arm B live: the reviewer's confound is real, not hypothetical. Section 6.5 states the confound explicitly and pre-registers the severe follow-up (generator regularization confined to train support, tested on a novel group element or modulus) with kill criteria. Until it runs, R2's strongest defensible reading of E4 is "aligned train-time intervention recovers external OOD where readout selection does not," with the mechanism not yet isolated.

R3. "The anti-Goodhart control loop reads like a philosophical compression, not an empirical result."

Response. We ran the named severe test rather than retaining the philosophical compression. The timestamped addendum froze all eight `allocate × cool × reopen` cells, eight paired seeds, existing terminal success, main and interaction contrasts, and transported controls before implementation. The strict verdict is **FAIL**. All-on passes 8/8 and removing reopen fails 8/8 (final MAE 0.113 → 0.550; reopen ratio 16.667 → 0), but removing allocate or cool still passes 8/8. Terminal main effects are allocate 0.0, cool 0.0, reopen +1.0; all terminal interaction contrasts are 0.0. Allocation does matter below threshold — selectivity 17.188 vs 4.125 and 23.1 vs 27.1 probes with reopen on — but it is not necessary for the frozen criterion. The cooling ablation is behaviorally identical under this schedule. We therefore reject the strong `{allocate, cool, reopen}` load-bearing-subset claim and retain only the bounded claim that the explicit second-shift reopen gate is necessary in this finite policy harness. C1–C3 and C5 do not depend on M4.